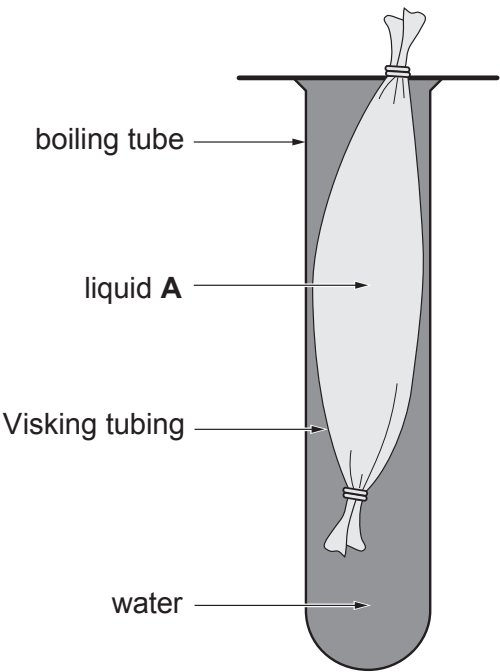


3. The cell membrane controls which substances enter and leave the cell. Visking tubing can be used as a model of the cell membrane.

The diagram shows the apparatus used at the start of an investigation using Visking tubing.
The contents of liquid **A** were tested for glucose and starch at the start of the investigation.
Liquid **A** and the water in the boiling tube were tested for glucose and starch at 30 minutes.



(a) (i) Complete the table below to show the results of tests made on liquid **A** at the **start of the investigation**. [2]

Test	Colour change	Conclusion
Benedict's reagent to test for glucose	blue to	glucose present
Iodine solution to test for starch	brown to	starch present

(ii) Describe how you would test the water in the boiling tube for the presence of glucose using Benedict's reagent.

(You do not need to describe the result) [1]

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(b) The table below shows the results for tests carried out on liquid **A** and the water in the boiling tube at 30 minutes.

Results at 30 minutes

	Test for glucose	Test for starch
liquid A	glucose present	starch present
water in boiling tube	glucose present	no starch present

Explain why, at 30 minutes, glucose molecules are present in the water in the boiling tube but starch molecules are not. [3]

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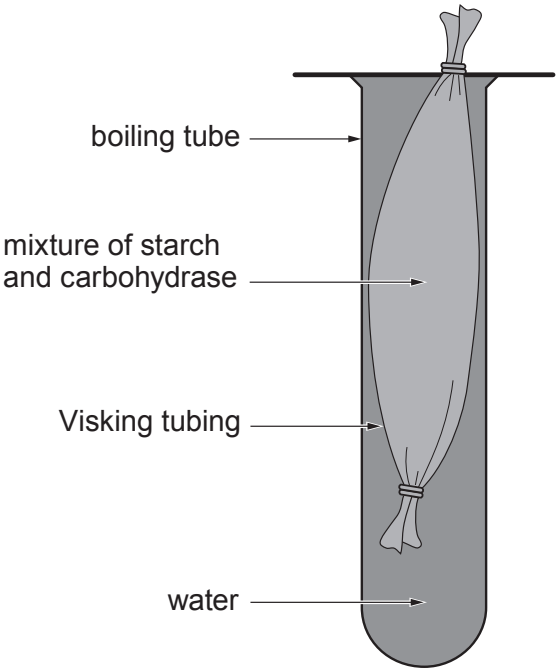
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(c) In a similar investigation, the Visking tubing contained **only** starch and the enzyme carbohydrase, as shown below.



The water in the boiling tube was tested at 30 minutes for the presence of glucose using Benedict's reagent.

Describe the result you would expect to observe. Explain your answer. [3]

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